

12

012-353

Monday

Zanzibar Revolution Day (Tanzania)

development is smooth & it is very popular.
open source JS Library

January 2015

JavaScript & JSX
Supports Mobile App.

Wk - 3

Zanzibar Revolution Day (Tanzania)

MERN STACK stands for MongoDB,

Express, React, Node, after the four key technologies that make up the stack.

- Mongo DB → document database
- Express (.js) → ^{middleware} Node.js web framework
- React (.js) → client-side JavaScript framework.
- Node (.js) → The premier JavaScript web server.

MEAN STACK : { JavaScript framework
use type script lang.
Does not support Mobile App.

M → Mongo DB

document db

E → Express JS

(backend web framework)

A → Angular JS

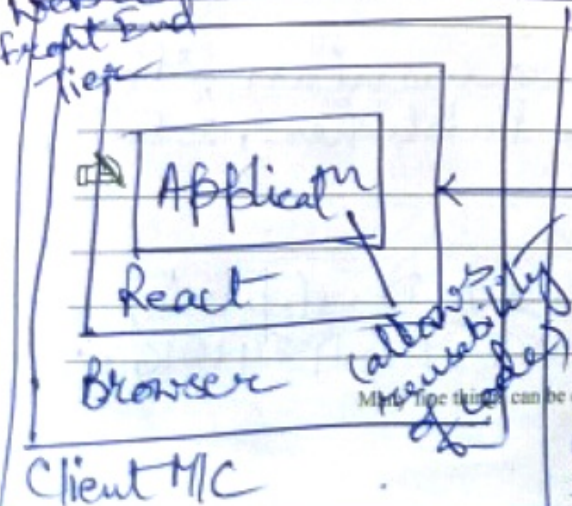
(front end framework)

N → Node JS

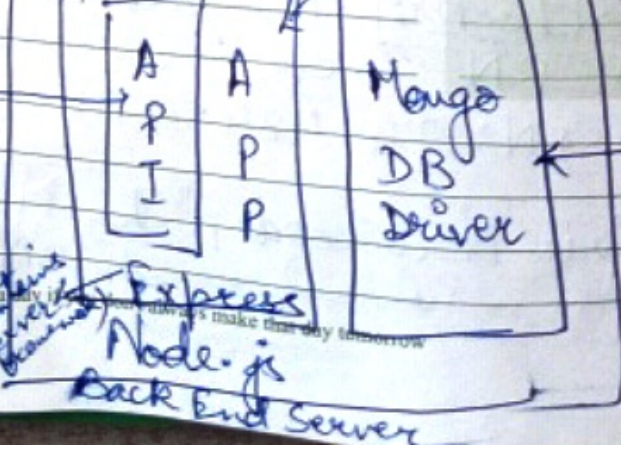
(backend runtime environment)

JSON → JavaScript Object Notation Format

Web or
Front End
Tier



Source or
Middle Tier



DB as
Backend
Tier

Mongo
DB

Many things can be done in a day if you have the right tools. Developers make this day tomorrow.

January 2015

Properties
(values)

013-352

Tuesday

13

CSS: Cascading Style Sheet

Types

- (i) Internal: holds CSS rules for the page in the head section of the HTML File.
- (ii) External: It is a file containing only CSS syntax.
- (iii) Inline CSS: allows you to apply a unique style to one HTML element at a time.

React: Visuals
Informations
Data fetching
Data display

is
Image
CSS } static
files

14

014-351

Wednesday

Makar Sankranti / Pongal (India, Sri Lanka)

Node / Nodepad++ /
pad visual studio

January 2015

Wk - 3

Appointments

JavaScript : is the most popular language in the world for the web.

we can also call it scripting language

→ There are two types of scripting language:

- ↳ (i) Client side scripting language ^{eg java script}
- (ii) Server side scripting language ^{eg: PHP, .Net}

Uses of JavaScript:

- ↳ Web Development
- ↳ App. Development
- ↳ Software Development
- ↳ Game Development etc.

Express JS setup → install

Node JS setup → install

React JS setup → install

Angular JS setup → install

Visual Studio

❖ Mongo DB setup → install : add / connect Mongo DB as VS code extensⁿ.

↓
(from VS Code).

If you tell the truth you don't have to remember anything. Mark Twain

January 2015

HTML → creates the structure of web page
CSS → it makes the design (layout) of web page
JavaScript → it works on behaviour of web page.
jQuery & React are the library of JS.

Appointments

Thursday

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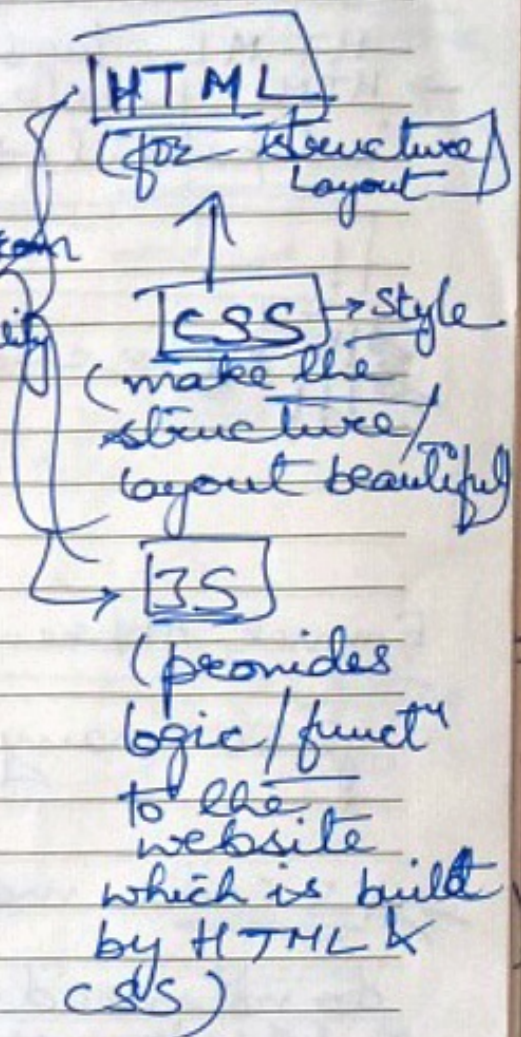
(Marquee, hyperlink tag, how to put image, audio & video)

Floccula

Ultra Edit (paid & w)

Visual Studio Code is freely available from
VS Code Editor → External functionality
(It is a notebook)

It also written in notepad



→ To see the page source code of any website
↓
right click

to view the page source

→ To see the source code in much clean format
↓
right click

Inspect



Structure

HTML space
221 2

How will you customise your web page

Elements Console Sources
↓
Devices

Dimensions: iPhone, Samsung galaxy, iPad mini
↓
Size of page

Happiness is not the absence of problems but the ability to deal with them

16 Friday

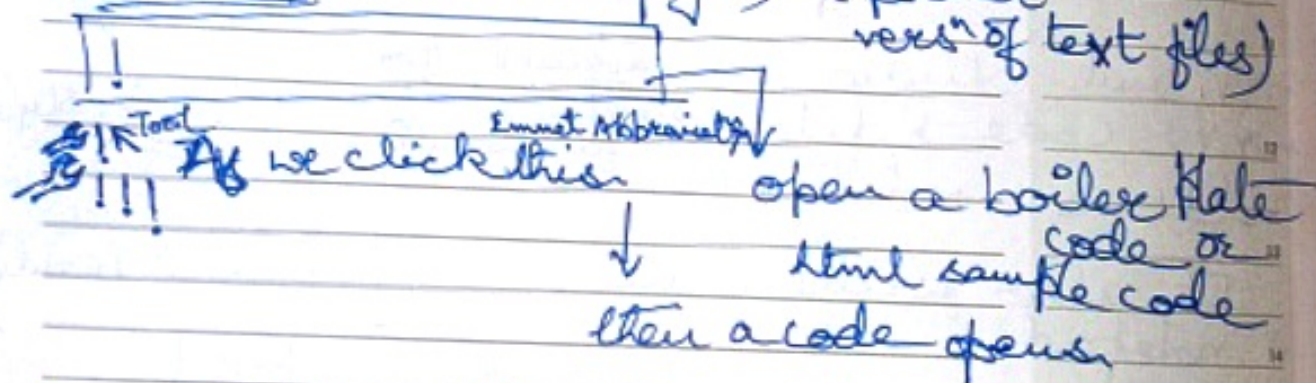
016-349

HTML document has 2 sections: (i) Head (ii) Body

January 2015

HTML: is the code that is used to structure a web page & its contents.

- The components used to design the structure of websites are called HTML tags.
- HTML files ends with .html or .htm extension.
index.html → (home page) (updated version of text files)



Emmet Abbreviations: it a ~~tot~~ tool kit, it gives many helpful tools.

→ VS Code makes it easy that, you do not need to learn, see & write the whole code. This functionality is not available on notepad.

We have also installed an extension live server on VS Code.

January 2015

Text files → .txt
Python files → .py

017-348

Saturday

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Appointments

Wk - 3

You will write the ^{html} code
on VS code Editor

↓
Save the Code

↓
Go Line of ^{html} at left bottom

↓
Then, you can see your
html page o/p

HTML Tag

: A container for some
content or other HTML tags.

<p> This is a paragraph </p>

Content

opening
tag

018-347

Sunday

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Closing
tag

Wk - 3

HTML Element

In VS code, we do not need to write
closing tags, they come automatically
with opening tag here.

To analyse

Basic HTML Page in VS Code

January 2015

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Monday

Chrome / Firefox, Safari

March 1st (US)

`<!DOCTYPE html>` tells browser that you are using HTML5

`<html>` root of an html document
`<head>` container of Meta Data
`<title>` My 1st Page `</title>` Page title
`</head>`

`<body>` contains the data which is rendered by browser
`<p>` Hello world! `</p>` paragraph tag
`</body>`

`</html>`

`<!DOCTYPE html>`

`<html lang="en">` → lang attribute specifies the language of the element's content.
"en" → English, "es" → Spanish, "fr" → French

`<head>`

`<meta charset="UTF-8">`

`<meta name="viewport" content="width=device-width, initial-scale=1.0">`

`<title>` Document `</title>`

`</head>`

`<body>`

`<p>` This is a paragraph `</p>`

`<p>` --- `</p>`

`</body>`

`</html>`

The common question that gets asked in business is, 'why?' That's a good question, but an equally valid question is, 'why not?' - Jeff Bezos

browser takes the html code & converts it into web page or website.

January 2015

browser $\frac{2}{\text{html}}$ $\frac{1}{\text{web page}}$ $\frac{1}{\text{render}}$ $\frac{1}{\text{Karana web page}}$ $\frac{1}{\text{ki form me}}$ $\frac{1}{\text{Tuesday}}$ **20**

Appointments

Tag/
Head/

that tells the information which do not be displayed on web page but must be known by the code.

Body Tag/

tells the information which actually displays on the web page.

(HTML5 specification)

→ This attribute encourages web developers to use UTF-8 character set, which covers almost all of the characters & symbols in the world!

<head>

→ MDN Reference (Developer's guide) shows the content of the particular word selected

→ viewport means website must be responsive. (As we change the screen size or aspect ratio then we see how much the particular website is responsive)

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021-344

Wednesday

Quick Points about HTML Wk - 4

Appointments

- HTML tag is ^{the} parent of head & body tag.
- Most of HTML elements have opening & closing tags with content in b/w.
- Some tags have no content in b/w, eg. ~~
~~ [↳] break line tag.
- we can use inspect element / view page source to edit HTML.

If To edit any attribute on web page then ^{right} click on HTML web page

↓
view source code
↓
or simply inspect to edit anything.



January 2015

 → non breaking space
entity

022-343

Thursday 22

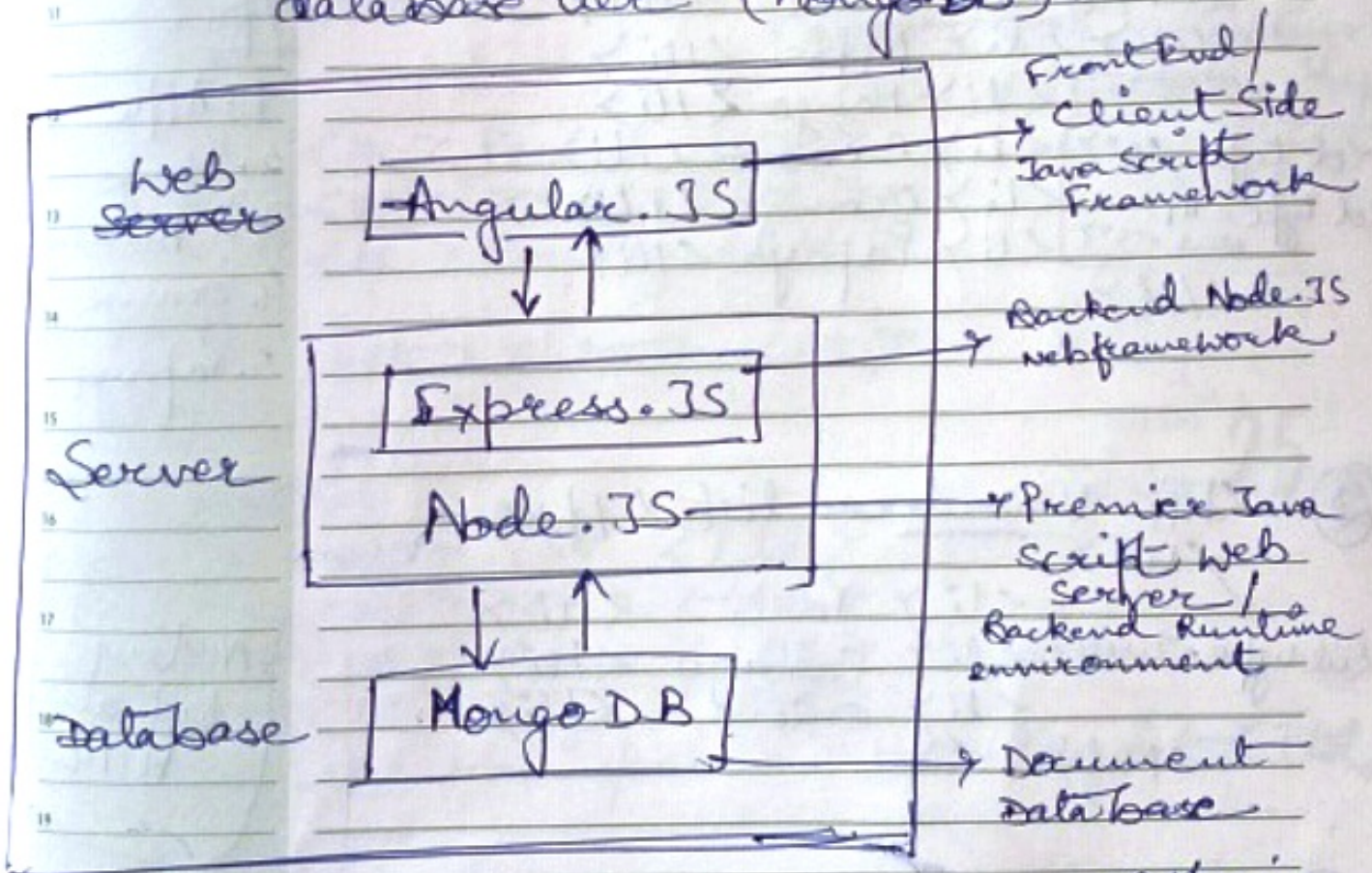
Wk - 4

Appointments

JavaScript Based Framework

Mean Stack : follows the traditional

three-tier stack pattern, including
the display tier (Angular.js),
application tier (Express.js &
Node.js),
database tier (MongoDB).



Variants of MEAN Stack :

- **MERN** (Replacing Angular.js with React.js)
- **MEVN** (using Vue.js)

23 Friday

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January 2015

Wk - 4

List in HTML 3 Types

- (i) ordered
- (ii) Unordered
- (iii) Definition

① <body>

<h1> Ordered List </h1>

 Apple

 Mango

 Banana

 Orange

 Papaya

> <ol type="A">

> <ol type="I" start="5">

o/p.:

1. Apple

2. Mango

3. Banana

4. Orange

5. Papaya

② <h1> Unordered list </h1>

 Potato

<ul type="circle"> Cabbage

 Onion

③ <h1> Definition List </h1>

<dl>

<dt> Rohit </dt>

<dd> Rohit is a good cr Player </dd>

</dl>

Nesting
List in
HTML

January 2015

Rowspan attribute → to merge the rows
Colspan attribute → to merge the column

024-341

24

Saturday

<tr>

↓
table row

<th>

table
heading

<td>

data in
table

Table in HTML

<!DOCTYPE html>

<html>

<head>

<title> Time Table </title>

</head>

<body>

<h1> School T T </h1>

<table border = "1px">

width = "60%" cellpadding = "10px" cellspacing = "0px"

<tr>

<th> Period </th>

<th> Time </th>

<th> Monday </th>

<th> Saturday </th>

</tr>

<tr>

<td> 1st </td>

<td> 10-11 </td>

<td> Chem </td>

<td> Eng. </td>

<td rowspan = "3"> Project </td>

</table>

<body>

<tr>

<td colspan = "8"> Lunch Break </td>

</table>

</body>

025-340

25

Sunday

Whatever the mind of man can conceive and believe, it can achieve. - Napoleon Hill

26

026-339

Monday

Wk - 5

Republic Day (India), National Day (Australia)

FORM In HTML:

+ Save

.html

XHTML:

(Extensible Hyper Text Mark-up Language)

- Similar to HTML
- More strict than HTML.

Major Differences from HTML

.xml version

- (i) DOCTYPE is mandatory
- (ii) `<html>`, `<head>`, `<title>` & `<body>` are mandatory.
- (iii) Elements must be properly nested & always be closed.
- (iv) Lowercase, one root element.

`<title>``</title>``<body>``</body>`# Steps to convert from HTML to XHTML.

- (i) Add XHTML `<!DOCTYPE>` to the first line of every page.
- (ii) Add `xmlns` attribute to `html` elements.
- (iii) Change all element name to lowercase.
- (iv) Close all elements.
- (v) Change all attribute names to lowercase.

Shift + pad. start

January 2015

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XHTML Document Structure

Tuesday

Appointments

✓ `<?xml version="1.0">`

✓ `<!DOCTYPE HTML PUBLIC "-//W3C//DTD`

XHTML 1.1//EN" `http://www.w3.org/`

`TR/xhtml11/DTD/xhtml11.dtd>`

(document type definition)

✓ `<html xmlns="http://www.w3.org/1999/xhtml">`

`<head>`

`<title>` Title of a web Page `</title>`

`</head>`

`<body>`

Contents to be displayed on the web page.

`</body>`

`</html>`

28

028-337

Wednesday

UTFUniversal Text Format - 8
Encoding

January 2015

Wk - 5

Every XHTML Document must begin with

<?xml version="1.0" encoding="UTF-8"?

<!DOCTYPE html PUBLIC "-//W3C//DTD

XHTML 1.1//EN" http://www.w3.org/TR

xhtml11/DTD/xhtml11.dtd>

• Doc type

Three types

- * Strict
- * Transitional
- * Frameset

//

- Multimedia Systems
- New Computing
- MultiMedia Tools

Q

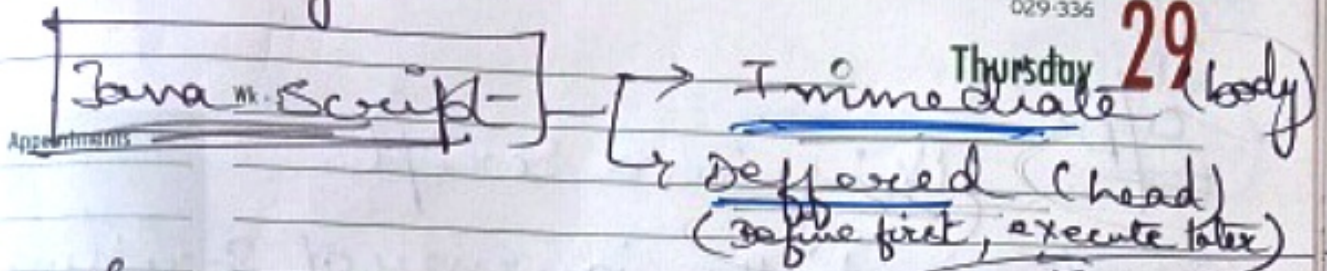
January 2015

Sublime → javascript.
jst.html

029-336

29

Thursday



Example:

```
<html>
<head>
<title>
```

Testing Java Script

```
</title>
```

```
</head> <script type="text/javascript"
function test() // Deferred Scripting
```

```
{
  alert document.write("<h3>
  Thanks for visiting </h3>");
}</script> </head>
```

```
<body>
```

```
<h2> Using JavaScript </h2>
```

```
<script language="JavaScript">
```

```
document.write("<b> welcome to the
journey of scripting </b>");
</script>
```

```
<h3> Printing my name 5 Times </h3>
```

```
<script type="text/javascript">
```

```
var a;
for (a=1; a<=5; a++) // Immediate Scripting
```

```
{
  document.write("<sup>kritika</sup>
  Anyali <br/>");
}
```

```
test();
</script>
```

```
</body>
```

```
</html>
```

The only person you are destined to become is the person you decide to be. - Ralph Waldo Emerson

30 030-335 Friday

O/p:

Wk - 5

Using Java Script

welcome to the journey of scripting

Printing my name 5 Times

Kritika

Kritika

Kritika

Kritika

Kritika

Appointments

To run the JS code

(1) Go to terminal → New Terminal

PS C:\Users\91989\Desktop\javascript>
node filename

→ Then enter → Then enter

| | | | | | |
|-------------|------|--------|------|----|-----|
| File | Edit | Select | View | Go | Run |
| ... | | | | | |
| Terminal | | | | | |
| JS index.js | | | | | |

Jan / Feb 2015

Node JS

To know
in ~~out~~ Go
031-
to
Saturday 31

\$.console.log('tech abc')

alert('tech abc') → shows alert box

Prob. o/p Debug Console Terminal

PS C:\Users\asus\91989\Desktop\javascript -> node
index.js
(file name)

o/p: Tech abc

Install Node JS

↳ Search Node JS in Google

032-333

Sunday

↳ 1st link Download for Windows 64

Win - 64

Abolition of Slavery Day (Monday)

✓ 14.16.1 LTS
Recommended
for Most Users

16.0.0 Current
latest Features

→ Then open Visual Studio Code

✗ File → open Folder → Desktop

Make a
file in
JavaScript
folder

← (JavaScript)
Folder
Name

make New
folder

index.js
(file)

Monthly Planner

February
2

| | | | |
|----|----|-------------------------------------|------------------------------|
| Su | 1 | Console.log("Hello everyone"); | |
| Mo | 2 | </script> | Comment |
| Tu | 3 | var a = 30; | /* --- |
| We | 4 | var b = 40; | --- |
| Th | 5 | var c = a + b; | --- |
| Fr | 6 | Console.log(c); | */ |
| Sa | 7 | | |
| Su | 8 | </script> | // ----- Single line Comment |
| Mo | 9 | | |
| Tu | 10 | → Pure JS → need to install node JS | |
| We | 11 | for HTML Coding | |
| Th | 12 | (4 Extensions) | |
| Fr | 13 | → (i) Code Runner | |
| Sa | 14 | → (ii) JavaScript (ES6) Code | |
| Su | 15 | | |
| Mo | 16 | → (iii) Live Server | |
| Tu | 17 | | |
| We | 18 | (iv) Prettier Code | |
| Th | 19 | | |
| Fr | 20 | | |
| Sa | 21 | | |
| Su | 22 | | |
| Mo | 23 | | |
| Tu | 24 | | |
| We | 25 | | |
| Th | 26 | | |
| Fr | 27 | | |
| Sa | 28 | | |

February 2015

Live Script after 1995 → JavaScript is invented
(by Netscape Corporation)
Java + Script
038-332

Monday

2

Condensation Day (Purification of Virgin Mary)

JavaScript

Appointments

client side, event based

- 1) It is an interpreted programming language with object oriented capabilities.
- 2) It is a dynamic programming lang.
(core concepts of JavaScript)
Multiple inheritance does not support in JavaScript
- 3) client-side scripting
- 4) Browser → JavaScript Engine.
(JavaScript programs are run by an interpreter built into the user's web browser.)
- 5) It is case sensitive, JS ignores white space.
You can use white space in your program.
- 6) JavaScript is not Java.

What can JavaScript do?

- 1) It can ^{dynamically} modify an HTML Page.
- 2) It can validate user i/p
- 3) It can be used to create cookies.
- 4) It is a full featured programming language.

3

034-331

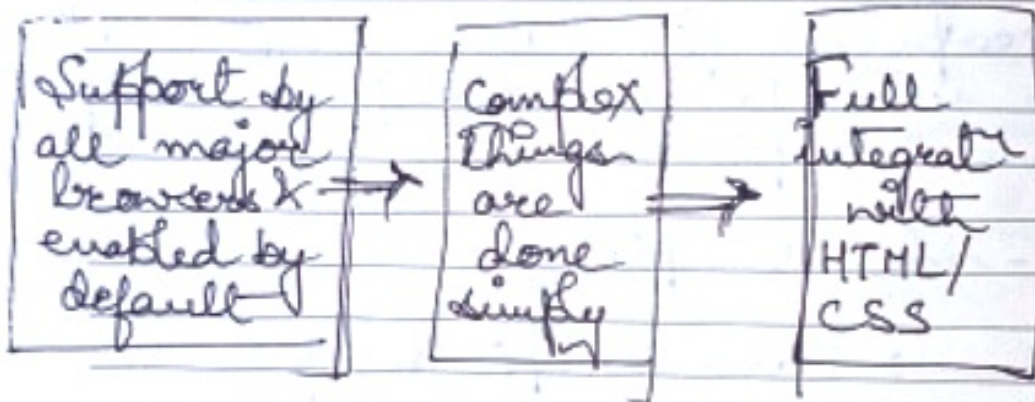
Tuesday

Full Moon

February 2015

5) JS user interaction does not require any communication b/w with the server.

What makes JavaScript unique?



JavaScript Frameworks & Libraries

| | | |
|------------------------|---|----------------------|
| 1. Angular (Framework) | J | 2. React (Framework) |
| | A | |
| 3. jQuery (Library) | V | 4. Vue.js |
| | A | |
| 5. Ext.js | S | 6. Ember.js |
| | C | |
| 7. Node.js | R | 8. Polymer |
| | I | |
| 9. Aurelia | P | 10. Backbone.js |
| | T | |

February 2015

<script>

</script>

035-330

Wednesday

4

Basic Syntax of Java Script

→ It is of two types

Type-I

<script type = "text / javascript" (optional)

document.write ("Education for you");

</script> And <script>

alert ("Welcome to JS");

</script>

Type-II

<script language = "Javascript">

document.write ("Education for you");

</script>

The script tag contains two important attributes

→ language: This attribute specifies what language you are using. Typically its value is Javascript.

→ Type: It indicates which script language you are using with type attribute.

5

036-329

Semicolons:

→ In JavaScript, semicolons terminate the statement. It is optional. For good programming you should use semicolons.

Comments: It is the same like as in C++ and any other languages.

JavaScript: Two types of Pop-up boxes
 (return true) (return false)

(i) Confirm Box → OK or cancel

(ii) Pop-up Box → OK

(iii) Prompt Box → OK or cancel

↓
the box returns the
input value

↓
the box returns
null.

Frames

Frames in HTML (make a folder)
 → Then makes html files

```

<!DOCTYPE html>
<html>
  <head>
    <title> Home Page </title>
  </head>
  <body>
    <h1> Home Page </h1>
  </body>
</html>

```



The best time to plant a tree was 20 years ago. The second best time is now. - Chinese Proverb

February 2015

037-328

Friday

6

Christchurch (New Zealand)

Frame Tag :- Frame set
frame Tag

HOME / ABOUT / CONTACT

Home Page

© Copyright 2024
Student / Courses / Development / Project

(i) we will make three pages for
HOME Page, About Page, Contact Page
Save home.html, about.html, contact.html
(Here, we will use anchor tag)

(ii) Make a new page (Header)

```
<!DOCTYPE html>  
<html>  
  <head>  
    <title>Header Page</title>  
  </head>
```

If you're offered a seat on a rocket ship, don't ask what seat! Just get on. - Sheryl Sandberg

7

038-327

Saturday

Wk - 6

Appointments

```

</body>
<h2 align="center">
<a href="home.html" target="framebody">
    Home </a>

```

```

<a href="about.html" target="framebody">
    About </a>

```

```

<a href="contact.html" target="framebody">
    Contact </a>

```

```

</h2>
</body>
</html>

```

8

039-326

Wk - 6

(iii) Make another page footer.

```

<!DOCTYPE html> | footer.html
<html>
  <head>
    <title> footer </title>
  </head>
  <body>
    <h2 align="center">
      * Copy; Copyright 2022 <br>
      Student / Courses / Development / Project
    </h2>
  </body>
</html>

```


(iv) Add one more page

Appointments

MAIN Page

(In which, we embed all 5 pages)

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title> MAIN PAGE </title>
```

```
</head>
```

```
<frameset rows="15%, 70%, 15%">
```

Child
tag
of frames

```
<frame name="frameheader" src="header.  
html"></frame>
```

```
<frame name="framebody" src="home.  
html"></frame>
```

```
<frame name="framefooter" src="footer.  
html">
```

```
</frame>
```

```
</frameset>
```

```
</html>
```


I - Frame

Wk - 7

Appointments

(Improved version of frameset)

<head>

<title> I Frame Index </title>

</head>

<body>

<header>

<nav>

<menu align = "center">

<h2>

HOME

ABOUT

CONTACT

</h2>

</menu>

</nav>

</header>

Dedication of purpose is the starting point of all achievement. - W. Clement Stone

February 2015

→ for content of website you will use section tag

042-323

Wednesday

11

<section>
Appointments

```
<iframe name = "framebody" src =  
"home.html" width = "1200"  
height = "400" >  
</iframe>  
</section>
```

```
<footer align = "center" >  
<h2>  
<h2 align = "center" >
```

```
&copy; ; copyright 2024 <br>
```

```
Student | Courses | Development | Project  
</h2>  
</footer>
```

Header

HOME | ABOUT | CONTACT

About us

Body

Footer

Challenges are what make life interesting and overcoming them is what makes life meaningful. - Joshua J. Marine

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variables, Data, Data Types, Literals

February 2015

12

043-322

Thursday

Let's Quarter

Vol - 7

Appointments

JavaScript

1. console.log("Hello World!");

↳ (always expect a string value)

console.log(type of "Hello World!");

• Numbers → 7, 72, 42.5

• String → "abc 123"

• Boolean → True / False

• Null / Undefined

↓
(no value)

↓
(variable define karta hai par usme koi value nahi hai)

• BigInt → 123n

• Symbol }

console.log(2.234)

console.log(type of 2.234)

my JS is

O/P

Hello World!

String

array
object

O/P 2.234

number

true
boolean

123n

BigInt

Literals

Program mai direct value ko use karne ko literal bolte hai.

February 2015

Rohit Sharma; rohit.Sharma

044-321

Friday

13

Console.log(null)

Appointments

null
(grey muted type text)

Console.log(234 + 456)

literal value

690

Console.log("A" * "A")

NaN
(Not a number)

Variables

Var USD conversion rate
(keyword)

= 83.16

Var USD;

USD = 83.16

assignment operator

Variable declare

Initialise the variable value

Console.log(100 * USD)

8322

Console.log(1250 * USD)

104025

Console.log(199 * USD)

16560.78

Var USD

// variables name will start from -, letter, \$

// Var usdconversionrate → usdConversionRate

// Var usd-conversion-rate → usdConversionRate

Var USD;

Console.log(USD)

USD = 83.16;

o/p

Undefined

Laughter is the shortest distance between two people. - Victor Berge

14

045-320

Saturday

FORMS

in HTML

Name:

Name

(water mark)

Father Name:

Father Name

Mother Name:

Mother Name

Email:

Email

Gender:

☐ Male☐ Female

Phone Number:

Phone Number

15

046-319

Sunday

Date of Birth:

mm/dd/yy

Submit

<html>

Wk - 8

Account Name

<head>

<title> HTML FORM </title>

</head>

<body>

& Name:

<input type="text" name="Name" size="15"

maxlength="30" placeholder="Name">

Father Name:

<input type="text" name="Father Name" size="15"

maxlength="30" placeholder="Father Name">

Mother Name:

Email:

<input type="email" name="email" size="15"

maxlength="30" placeholder="Email">

Gender:

<input type="radio" name="gender" size="15"

value="Male"

maxlength="30">

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048-317

Tuesday

Rita Sharma (Info, Review)

Wk - 8

`<input type="radio" name="gender" value="female" size="15" maxlength="30" /> Female <br><br>`

Phone Number:

`<input type="text" name="text" size="15" maxlength="30" placeholder="Phone number" /> <br><br>`

Date of birth:

`<input type="date" name="date of birth" size="15" maxlength="30" /> <br><br>`

`<input type="submit" value="Submit" /> <br></body> </html>`

Appointments

Wk - 3

Function Syntax in JavaScript

```
function functionName()
{
    Statement
}
;
```

← Function Definition
(Function Declaration)

functionName(); ← Function Calling

Ex: <head>
 <title> Function of JS </title>
 <script>

Function Definition

```
function myIntro() {
    document.write('I am Shyam');
    document.write('<br>');
    document.write('I am an Animation Specialist');
    document.write('<br>');
}
```

Function Calling

```
myIntro();
myIntro();
```

</script>
 <body> </body> </head>

o/p:

I am Shyam
 I am an Animation Specialist

I am Shyam
 I am an Animation Specialist

Function with Parameter

```
function sum(a,b) {
  document.write(a+b);
}
```

```
sum(10,20);
```

```
...
```

```
sum(20,40);
```

or

```
function functionName (parameter1, parameter2) {
  Statement
}
```

Syntax

```
functionName (argument1, argument2);
```


Appointments

```
var a = 10;
var b = 20;
document.write(a+b);
```

```
...
...
```

```
var a = 30;
var b = 40;
document.write(a+b);
```

```
...
...
```

```
var a = 25;
var b = 55;
document.write(a+b);
```

★ Difference b/w console.log & document.write

↓
console.log
o/p show hoga
hai

↓
document.write
o/p show hoga hai

February 2015

21 ^{05:30} Saturday

{ } → object
[] → array

objects

Advance
Level of
Array

Wk - 1

Appointments

An object is a collection of properties & a property is an association b/w a name (or key) & a value. A property's value can be a functⁿ, in which case the property is known as a method. In addition to objects that are predefined in the browser, you can define your own object.

var fees = {

Rahul: 100,
Symit: 200,
Rohan: 300,

} → Properties

22 ^{05:31:2} Sunday

Total: function() { return (100+200+300); }

};

↓
Object

console.log(fees);

to document.write(fees.Rohan);

Method
↓
o/p
Rahul: 100
Symit: 200
Rohan: 300
300

Types of Objects: 4 Types

(i) User-defined objects - These are custom objects created by the programmer to bring structure & consistency to a particular programming task.

February 2015

054-311

Monday

23

- Imp
- (ii) Native objects: These are provided by the JS lang. itself like string, Number, Boolean, Function, Date, Object, Array, Math, RegExp, Error, as well as object that allow creation of user-defined objects & composite types.
- (iii) Host Object: These objects are not specified as part of the JS Lang. but that are supported by the host environment, typically browsers like window, navigator.
- (iv) Document Objects: These are part of the Document Object Model (DOM), as defined by the W3C. These objects present the programmer with a structured interface to HTML & XML documents. Access to the document objects is provided by the browser via the document property of the window object (window.document).

24 Tuesday

055-310



WK - 9

Appointments

Document Object Model: It is a

- Application Programming Interface (API) for HTML & XML documents.
- With the DOM, programmers can create & build documents, navigate their structure, & add, modify, or delete elements & content.
- DOM is a obj. oriented representation of the web page, which can be modified with a scripting lang. such as JS.
- DOM represents a document with a logical tree.
- Acc. to DOM, every HTML tag is an object. Nested tags are cfd "Children" of the enclosing one.
- All operations on the DOM start with the document object. From it we can access any node.
- The DOM can be used with any programming lang.

Appointments

Wk - 9

First Quarter

DOM Levels

- DOM Level 0: supports the common document obj. collect^{ns} - form[], image[], anchors[], links & applets[]. This is also known as classic or traditional JS Obj. Model.
- DOM Level 1: It provides the ability to manipulate all elements in a document through a common set of functⁿ. In this level all elements are exposed & parts of the page can be read & written to at all times.
- DOM Level 2: It provides further access to page elements primarily related to CSS & focuses on combining Level 0 & Level 1 while adding improved support for working with XML document.
- DOM Level 3: Enhanced Level of Level 1 & Level 2 & added support for XPath & keyboard event handling.
- DOM Level 4: Enhanced year 2015
↳ (latest DOM Level)

Categories of DOM

11-9

Appointments

- DOM Core: It specifies a generic model for viewing & manipulating a marked up document as a tree structure.
- DOM HTML: It specifies an extension of the core DOM for use with HTML. This represents DOM Level 0 with capabilities for manipulating all of the HTML element objects.
- DOM CSS: It specifies the interfaces necessary to manipulate CSS rules programmatically.
- DOM Events: It adds event handling to the DOM.
- DOM XML: It specifies an extension to the core DOM for use with XML.

DOM Tree

 Document
root (html)

 element: <head>
 element: <title>
 Text: 'New Website'

element: <body>

 element: <nav>
 element: <a>
 Text: attribute

 element: <h1>
 Text: 'Welcome'
 attribute

delete / Add / Get / Set / HTML element

Remember that not getting what you want is sometimes a wonderful stroke of luck. - Dalai Lama

Array in Java Script

Array: A Group of variables. An array can hold many values under a single name & you can access the values by referring to an index number.

Ex: Multiple variable declaration

```
<!doctype html>
```

```
<html>
```

```
<head>
```

```
<title> Array </title>
```

```
</head>
```

```
<body>
```

```
<script type = "text / javascript">
```

```
/* let sub1 = 'Math' ;
```

```
let sub2 = 'Phys' ;
```

```
let sub3 = 'Chem' ;
```

```
let sub4 = 'Bio' ;
```

```
document.write(sub1);
```

```
document.write(sub4);
```

O/p:

Maths

Bio

*/

Multiple variables ko agar
ek place par declare krke do
in use array kchta hai.

28 059-306 Saturday

var
let subject = new Array ('Math', 'Physics');

document.write (subject);

o/p:
Maths
Physics

New
way
Modern

let subject = ['Math', 'Physics'];

document.write (subject);

o/p:
Maths
Physics

060-305
let subject = ['Math', 'Physics', 'Chemistry', 'Bio'];

subject[3] = 'English';

document.write (subject[1]);

document.write (subject[3]);

o/p
Physics
English

Update

Sunday

</script>
</body>
</head>

`var a = 10;`

`var a = 10, 20, 30;` → Error

`var a = "10, 20, 30";` → String

`var a = [10, 20, 30];` → Array

0 1 2 ← index no.

↓ ↓ ↓

keyword variable name array starts from 0 index

`<!DOCTYPE html>`

`<html>`

`<head>`

`<title> Array </title>`

`<script>`

`var arry = [10, 20, 30, 40, 50];` ← index

0 1 2 3 4

o/p

`document.write(arry);` → 10, 20, 30, 40, 50

`document.write(arry[2]);` → 30

`document.write(arry[4]);` → 50

`document.write(arry[5]);` → Undefined

→ `for (var a = 0; a <= 4; a++)` o/p

{

`document.write(arry[a] + "<br>");` 10

} 20

30

40

50

`</script>`

`</head> <body> </body> </html>`

Monthly Planner

March

```

<script>
var arry = [10, 20, 30, 40, 50];
var sum = 0;
document.write("<ul>");
for (var a = 0; a < 5; a++)
{
    document.write("<li> + arry[a] + "</li>");
    sum = sum + arry[a];
}
document.write("</ul>");
document.write("Total Sum: " + sum);
    
```

• 10
 • 20
 • 30
 • 40
 • 50
 Total sum = 150

```

</script>
    
```

```

<script>
var arry = [10, "Harry", "Sarah", "true", null];
    
```

```

document.write("<ul>");
for (var a = 0; a <= 4; a++)
{
    document.write("<li> + arry[a] + "</li>");
}
document.write("</ul>");
    
```

Mixed
 data
 type
 can
 be
 added
 in
 one
 variable

```

</script>
    
```

o/p
 • 10
 • Harry
 • Sarah
 • true
 • Null

3

062-303

Tuesday

March 2015

Nested Array in Java Script

Appointment

other Array in array.

animals [cat | Dog | Hawk | eagle]

birds

0

1

2

animals = ["cat", "dog", ["hawk", "eagle"]]

Accessing
eagle

animals [2][1] → Eagle

animals [2][0] → Hawk

var cities = ["Delhi", "Mumbai",
"Chennai", ["a", "b", "c"]];

console.log(cities)

console.log(cities [3][1])

o/p

"Delhi"

"Mumbai"

"Chennai"

["a", "b", "c"]

b

5

064-301

Thursday

Full Moon

Wk - 10

Object

<body>

<script>

```
/* let person = ['Ishtak', 'Hussain', 40]
*/
```

```
let person = {
  first Name: 'Rohit',
  second Name: 'Sharma',
  age: 60;
};
```

```
console.log(person);
document.write(person.age);
```

</script>

</body>

o/p

```
first Name: Rohit
second Name: Sharma
age: 60
```

60

March 2015

065-300

Friday

6

Holi (India), Independence Day (Shona)

Diff b/w var, let, const

var Functional Scope

```
function() {  
  if (true) {  
    var a = 12;  
  }  
  console.log(a);  
}
```

Nearest function,
it is accessed any-
where

let Block Scope

```
function() {  
  if (true) {  
    let a = 12;  
  }  
  console.log(a);  
}
```

access
with in
this only
brace

const Block Scope

```
function() {  
  if (true) {  
    const a = 12;  
    a = 6; ✗  
  }  
}
```

access
with in
this (nearest)
brace

JavaScript Event Handling

Appointments

- Event Handling is a ~~the~~ routine that processes actions such as key stroke & mouse movements.
- It is the receipt of an event at some event handler from an event procedure & subsequent ~~procedure~~ processes.

Functions of Event Handling

- Event handling identifies where an event should be forwarded.
- It makes the forward event.
- It takes some kind of appropriate action in responses such as writing to a log, sending an error or recovery routine, or sending a message.
- The event handler may ultimately forward the event to an event consumer.

Event Handler

Description

On Abort

It executes when the user aborts loading an img.

Don't find fault, Find a remedy. - Henry Ford

March 2015

070-295

Wednesday

11

• On Blur

It executes when the
if focus leaves the
field of a text, textarea
or a select option.

OnChange → onclick, on error, on focus,
onload, on reset, on select, on submit,
onunload.

Simple program Onload() Event handler.

```
<html>
```

```
<head>
```

```
<script>
```

```
function type()
```

```
{  
    var d = new Date();
```

```
    var ty = d.getHours() + ":" + d.getMinutes()
```

```
    + ":" + d.getSeconds();
```

```
    document.getElementById("time").value = ty;
```

```
    SetInterval("time()", 1000)
```

```
}
```

```
</script>
```

```
</head>
```

```
<body> = on
```


12

071-294

Thursday

Independence / National Day (Kaziranga)

```

<body onload = "time()">
<center> <h2> Display Time </h2>
<form name = "form1">

```

```

<input type = text name = time text size = "8">

```

```

</form>

```

```

</center> </body> </html>

```

O/p:-

Display Time

18:45:30

Java Script Basic Events

- Click (onclick)
- Double click (ondblclick)
- Right click (oncontextmenu)
- Mouse Hover (onmouseover)
- Mouse Out (onmouseout)
- Mouse Down (onmousedown)
- Mouse Up (onmouseup)
- Key Press (onkeypress)
- Key Up (onkeyup)
- Load (onload)
- Unload (onunload)
- Resize (onresize)
- Scroll (onscroll)

It works only on body tag

Windows
event
(200)

Appointments

```

1 <html>
2   <head>
3     <title> JS </title>
4   <script>
5     function hello() {
6       document.write("Hello Everyone");
7     }
8   </script>
9 </head>
10 <body>
11   <button onclick="hello()"> Click Me </button>
12
13 </body>
14 </html>
15
16 <p onclick="hello()"> Click on Me </p>
17
18 <body onkeypress="hello()">
19
20

```

o/p
on web page

Click Me After
↓ clicking
Hello Everyone

Click me On

14 Saturday

Modifying Element Style in JS

Syntax:

```
document.getElementById(id).style.property
= new style
```

```
<html>
<body>
<h1 id="hid">ISS </h1>
```

```
<script>
```

```
document.getElementById("hid").style
```

```
color = "Red";
```

15 Sunday

```
</script>
```

```
</body> </html>
```

Events → clicking a button
 → I/p field is changed ...
 & so on.

March 2015

075-290

Monday

16

<button type="button" onclick =

Appointments

"document.getElementById(Aid).

style.color = "Red"> Click Me </button>

I don't like that man. I want get to know him better. - Abraham Lincoln

17

076-289

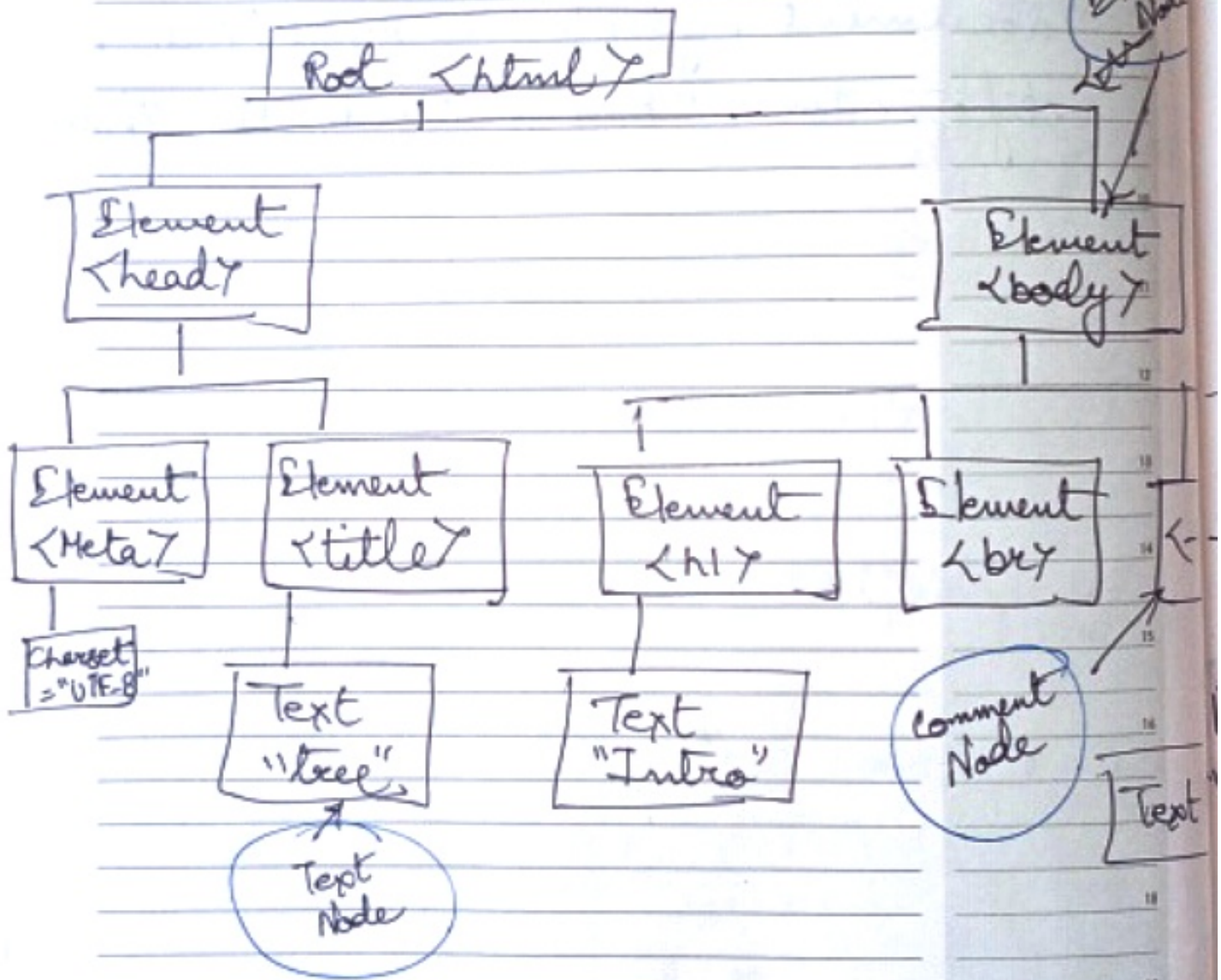
Tuesday

St. Patrick's Day (Northern Ireland, Republic of Ireland, USA)

Document Tree

Wk - 12

March 2015



March 2015

<html>
<head>
<meta charset="UTF-8">
<title> Tree </title>

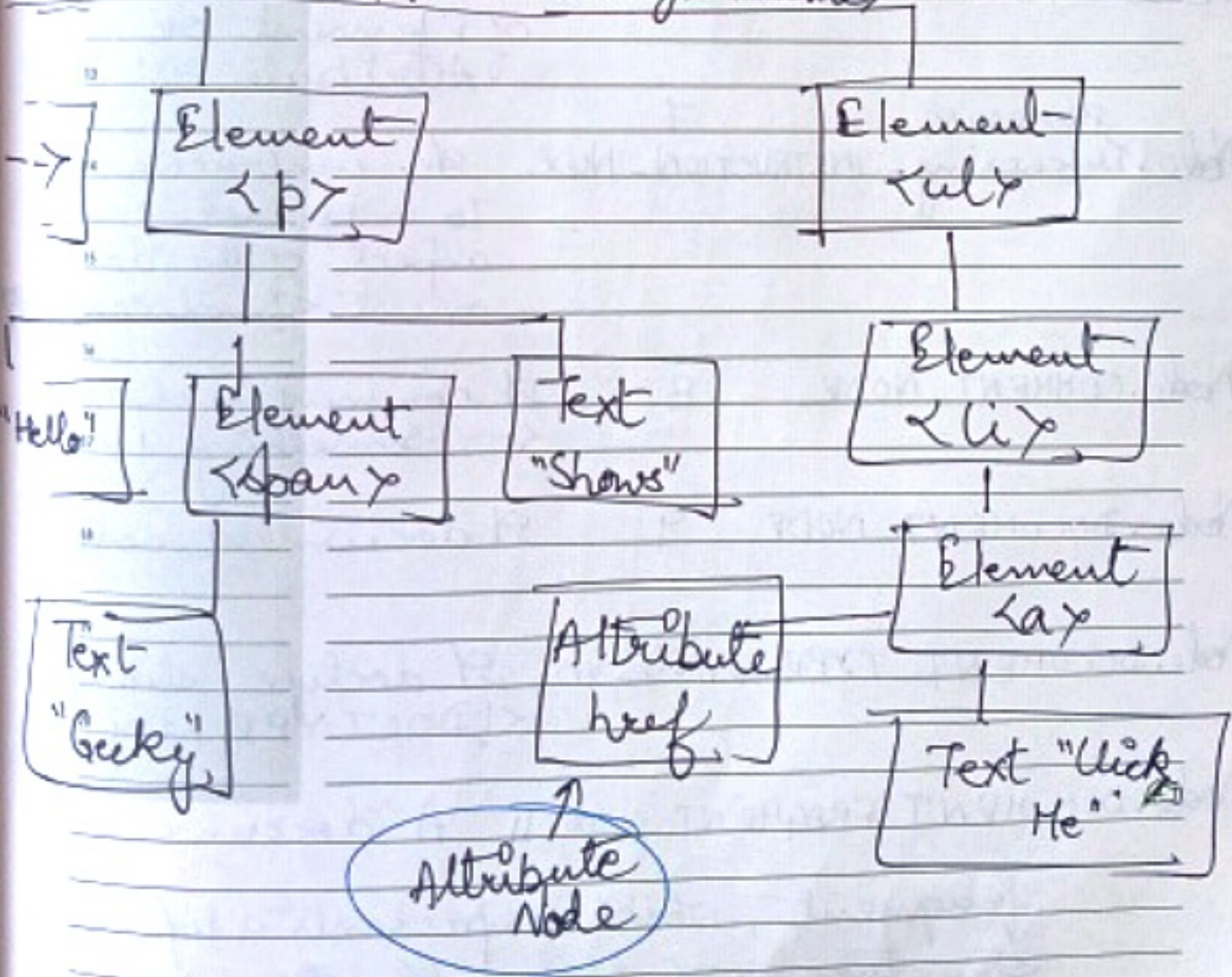
Appointments
<head>
<body>

<h1>Intro</h1>

<--- This is Para --->

<p>HelloGeekyShows</p>

</body></html> Click Me



To be wronged is nothing unless you continue to remember it. - Confucius

DOM Node Types

| <u>Constant</u> | <u>Node Type</u> | <u>Description</u> |
|--|------------------------------------|--|
| ^{ELEMENT}
Node.ELEMENT | NODE 1 | An element such as <code><h1></code> or <code><div></code> |
| ^{TEXT}
Node.TEXT | NODE 3 | The actual text of element or attribute ex:- "Hello" |
| ^{PROCESSING}
Node.Processing | ⁷
INSTRUCTION - NODE | An instruction to a parser on aspect of the document <code><?xml version="1.0"></code> |
| Node.COMMENT | NODE 8 | A comment such as <code><-- something --></code> |
| Node.DOCUMENT | NODE 9 | A document node |
| Node.DOCUMENT_TYPE | NODE 10 | A doctype statement <code><!DOCTYPE html></code> |
| Node.DOCUMENT_FRAGMENT | NODE 11 | A document fragment, which represents a lightweight structure to hold a collection of DOM nodes for manipulation or insertion. |

Rosencrantz is like taking poison and waiting for the other person to die. - Malachy McCourt

March 2015

079-286

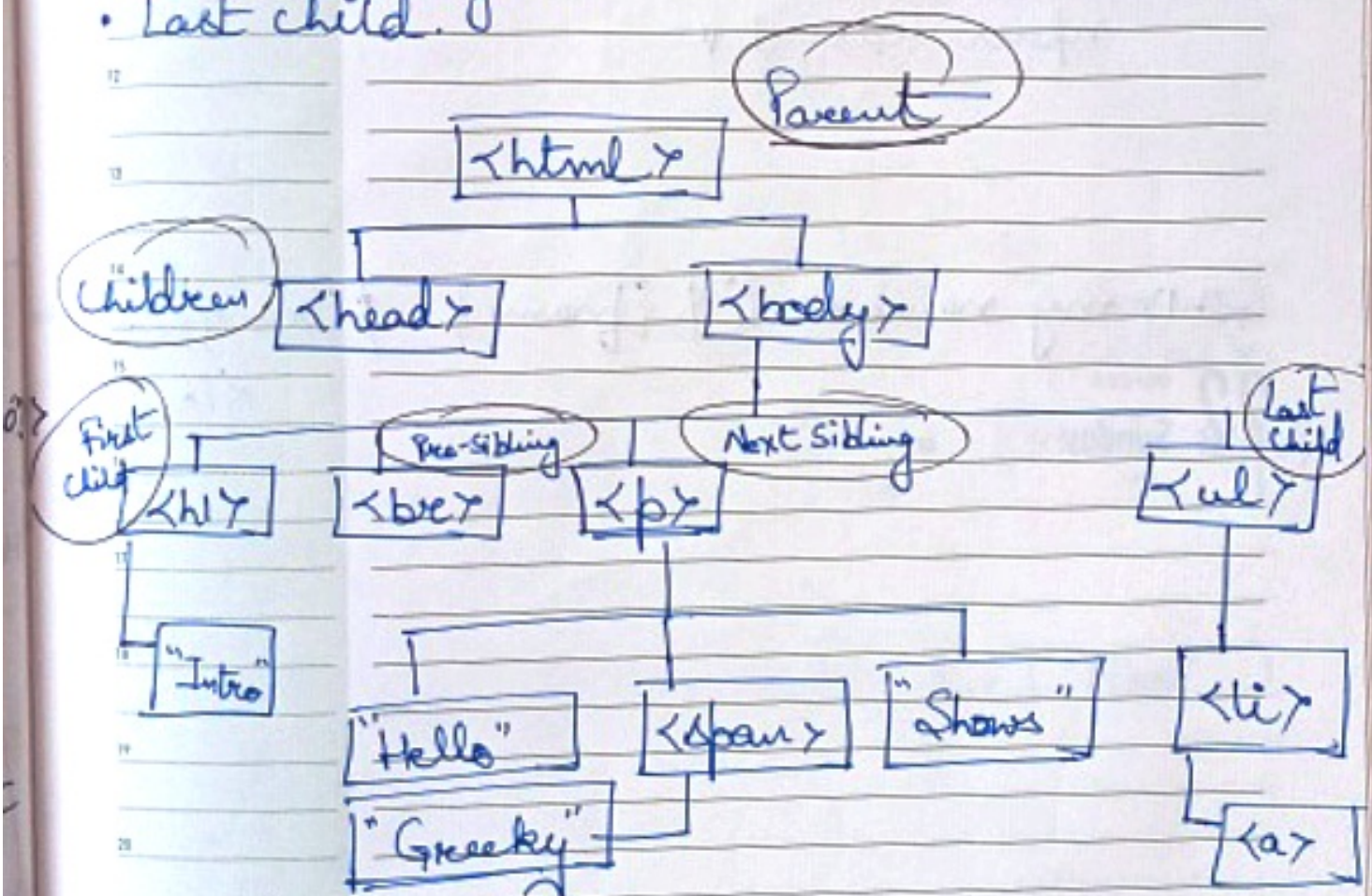
Friday 20

Appointments

Wk 12

Node Relationships

- Parent
- Children
- First child
- Previous sibling
- Next sibling
- Last child



March 2015

index.js

082-283

Monday

23

Template Literals

```
import React from "react";
import ReactDOM from "react-dom";
```

```
const fname = "Vinod";
```

```
const lname = "thapa";
```

```
ReactDOM.render(
```

```
<h1> my name is {fname + " " + lname} </h1>
```

```
<p> my lucky number is {5+5} </p>
```

```
</div>
```

```
document.getElementById("Root")
```

```
);
```